Advanced Javascript


Drum Kit
w
a
s
d
j
k
l



```html
<!DOCTYPE html>
<html lang="en" dir="ltr">

<head>
  <meta charset="utf-8">
  <title>Drum Kit</title>
  <link rel="stylesheet" href="styles.css">
  <link href="https://fonts.googleapis.com/css?family=Arvo" rel="stylesheet">
</head>

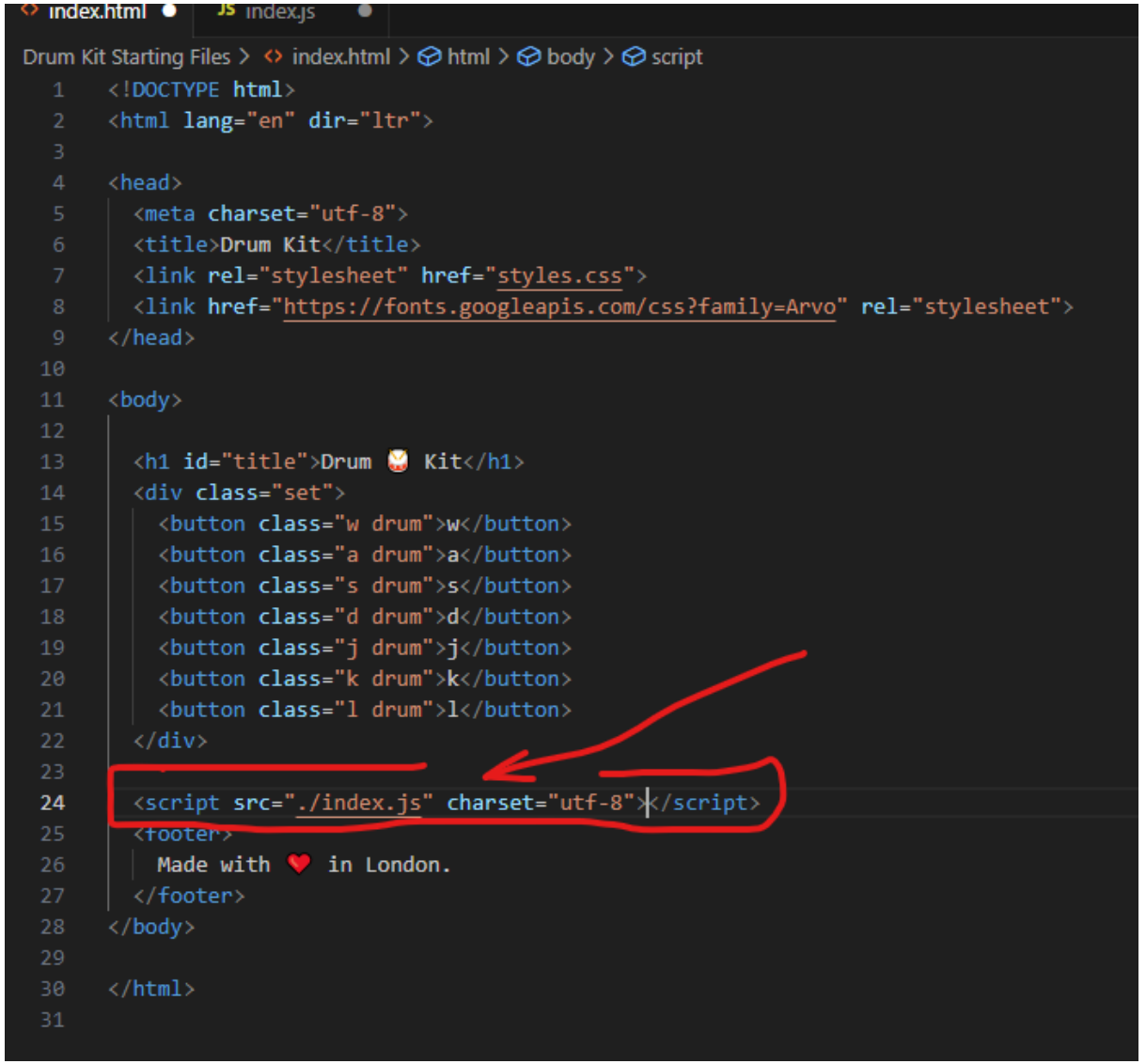
<body>
  <h1 id="title">Drum 🥁 Kit</h1>
  <div class="set">
    <button class="w drum">w</button>
    <button class="a drum">a</button>
    <button class="s drum">s</button>
    <button class="d drum">d</button>
    <button class="j drum">j</button>
    <button class="k drum">k</button>
    <button class="l drum">l</button>
  </div>



</body>
<footer>
  Made with ❤️ in London.
</footer>

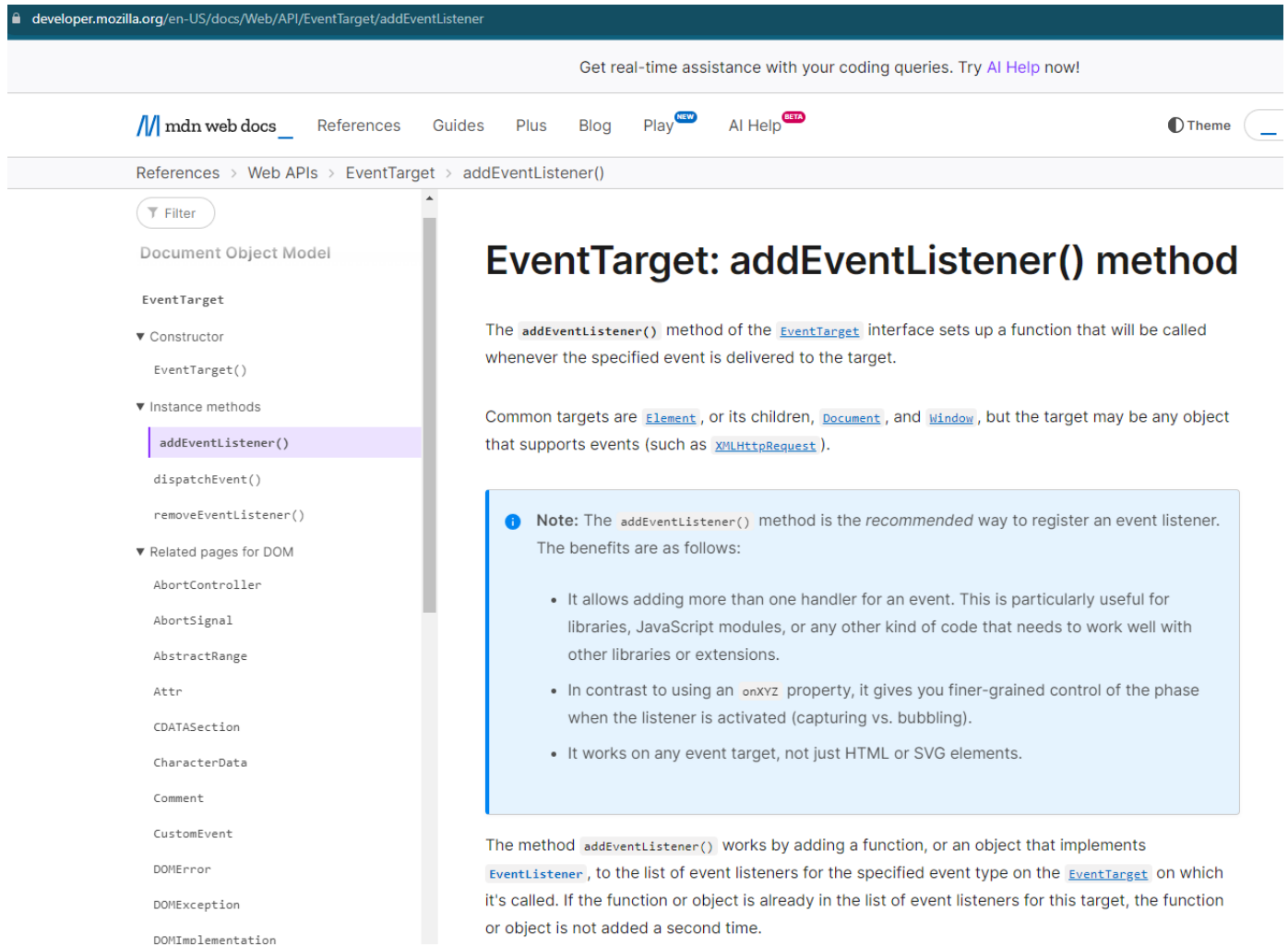
</html>
```

Push the script



Drum Kit Starting Files > JS index.js > handleClick

```
1   document.querySelector("button").addEventListener("click", handleClick);
2
3   function handleClick(){
4       alert("I got clicked!");
5   }
```


developer.mozilla.org/en-US/docs/Web/API/EventTarget/addEventListener
Get real-time assistance with your coding queries. Try AI Help now!
mdn web docs
References
Guides
Plus
Blog
Play
NEW
AI Help
BETA
Theme
References › Web APIs › EventTarget › addEventListener()
Filter
Document Object Model
EventTarget
Constructor
EventTarget()
Instance methods
addEventListener()
dispatchEvent()
removeEventListener()
Related pages for DOM
AbortController
AbortSignal
AbstractRange
Attr
CDATASection
CharacterData
Comment
CustomEvent
DOMError
DOMException
DOMImplementation
EventTarget: addEventListener() method
The addEventListener() method of the EventTarget interface sets up a function that will be called whenever the specified event is delivered to the target.
Common targets are Element, or its children, Document, and Window, but the target may be any object that supports events (such as XMLHttpRequest).
Note: The addEventListener() method is the recommended way to register an event listener. The benefits are as follows:
It allows adding more than one handler for an event. This is particularly useful for libraries, JavaScript modules, or any other kind of code that needs to work well with other libraries or extensions.
In contrast to using an onXYZ property, it gives you finer-grained control of the phase when the listener is activated (capturing vs. bubbling).
It works on any event target, not just HTML or SVG elements.
The method addEventListener() works by adding a function, or an object that implements EventListener, to the list of event listeners for the specified event type on the EventTarget on which it's called. If the function or object is already in the list of event listeners for this target, the function or object is not added a second time.

Syntax

JS

```
addEventListener(type, listener)
addEventListener(type, listener, options)
addEventListener(type, listener, useCapture)
```

Parameters

`type`

A case-sensitive string representing the event type to listen for.

`listener`

The object that receives a notification (an object that implements the `Event` interface) when an event of the specified type occurs. This must be `null`, an object with a `handleEvent()` method, or a JavaScript function. See The event listener callback for details on the callback itself.

`options` Optional

An object that specifies characteristics about the event listener. The available options are:

`capture` Optional

A boolean value indicating that events of this type will be dispatched to the registered `listener` before being dispatched to any `EventTarget` beneath it in the DOM tree. If not specified, defaults to `false`.

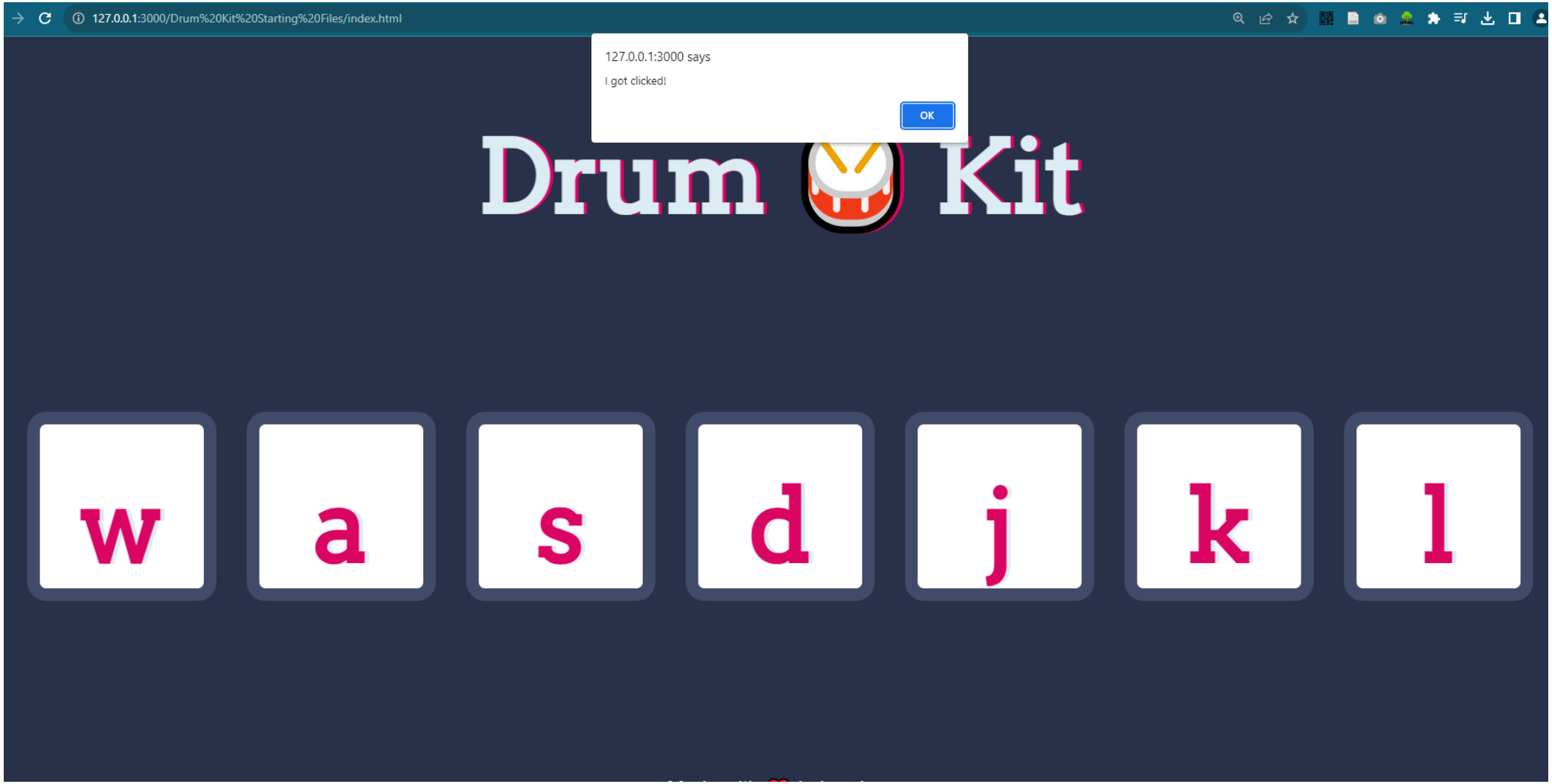
`once` Optional

Secure | https://developer.mozilla.org/en-US/docs/Web/Events

`keydown`	ANY key is pressed
`keypress`	ANY key except Shift, Fn, CapsLock is in pressed position. (Fired continously.)
`keyup`	ANY key is released

Mouse Events

Event Name	Fired When
`mouseenter`	A pointing device is moved onto the element that has the listener attached.
`mouseover`	A pointing device is moved onto the element that has the listener attached or onto one of
`mousemove`	A pointing device is moved over an element. (Fired continously as the mouse moves.)
`mousedown`	A pointing device button is pressed on an element.
`mouseup`	A pointing device button is released over an element.
`auxclick`	A pointing device button (ANY non-primary button) has been pressed and released on an e
`click`	A pointing device button (ANY button; soon to be primary button only) has been pressed a element.
`dblclick`	A pointing device button is clicked twice on an element.
`contextmenu`	The right button of the mouse is clicked (before the context menu is displayed).
`wheel`	A wheel button of a pointing device is rotated in any direction.


127.0.0.1:3000/Drum%20Kit%20Starting%20Files/index.html
127.0.0.1:3000 says
I got clicked!
OK
Drum Kit
w
a
s
d
j
k
l

Anonymous Function



Make a loop for all event

Drum Kit Starting Files > JS index.js > ...

```
1   var numberOfDrumButtons = document.querySelectorAll(".drum").length;
2
3   for(var i = 0;i< numberOfDrumButtons;i++)
4   document.querySelectorAll("button")[i].addEventListener("click", function (){
5       alert("I got clicked!");
6   });
7
8
```

Higher Order Function Challenge Solution

```
function add(num1, num2) {
return num1 + num2;
}

function subtract(num1, num2) {
return num1 - num2;
}

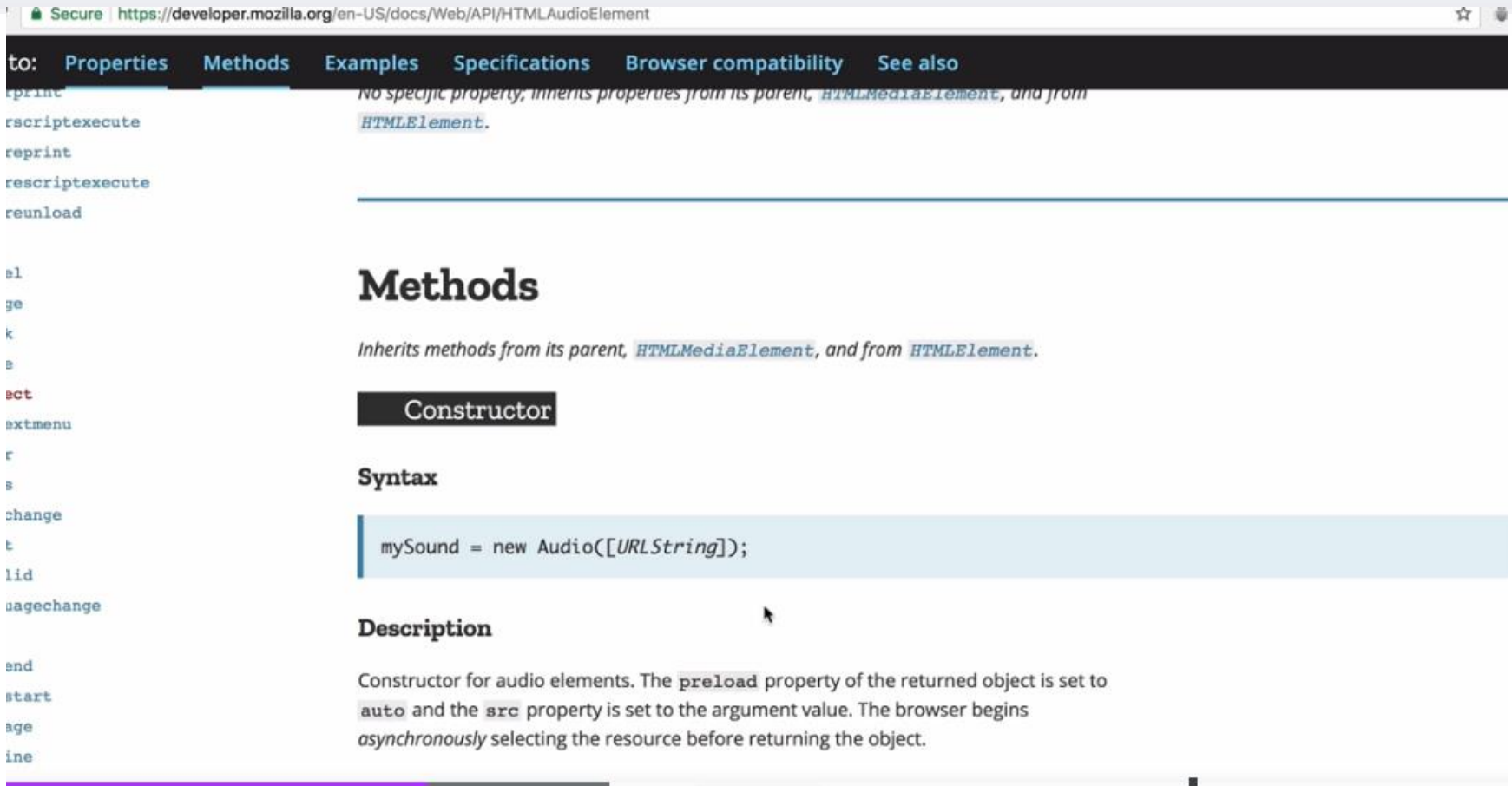
function multiply(num1, num2) {
return num1 * num2;
}

function divide(num1, num2) {
return num1 / num2;
}

function calculator(num1, num2, operator) {
return operator(num1, num2);
}
```

Play sounds on a website

```
var audio = new Audio('audio_file.mp3');
audio.play();
```



```
var numberOfDrumButtons = document.querySelectorAll(".drum").length;

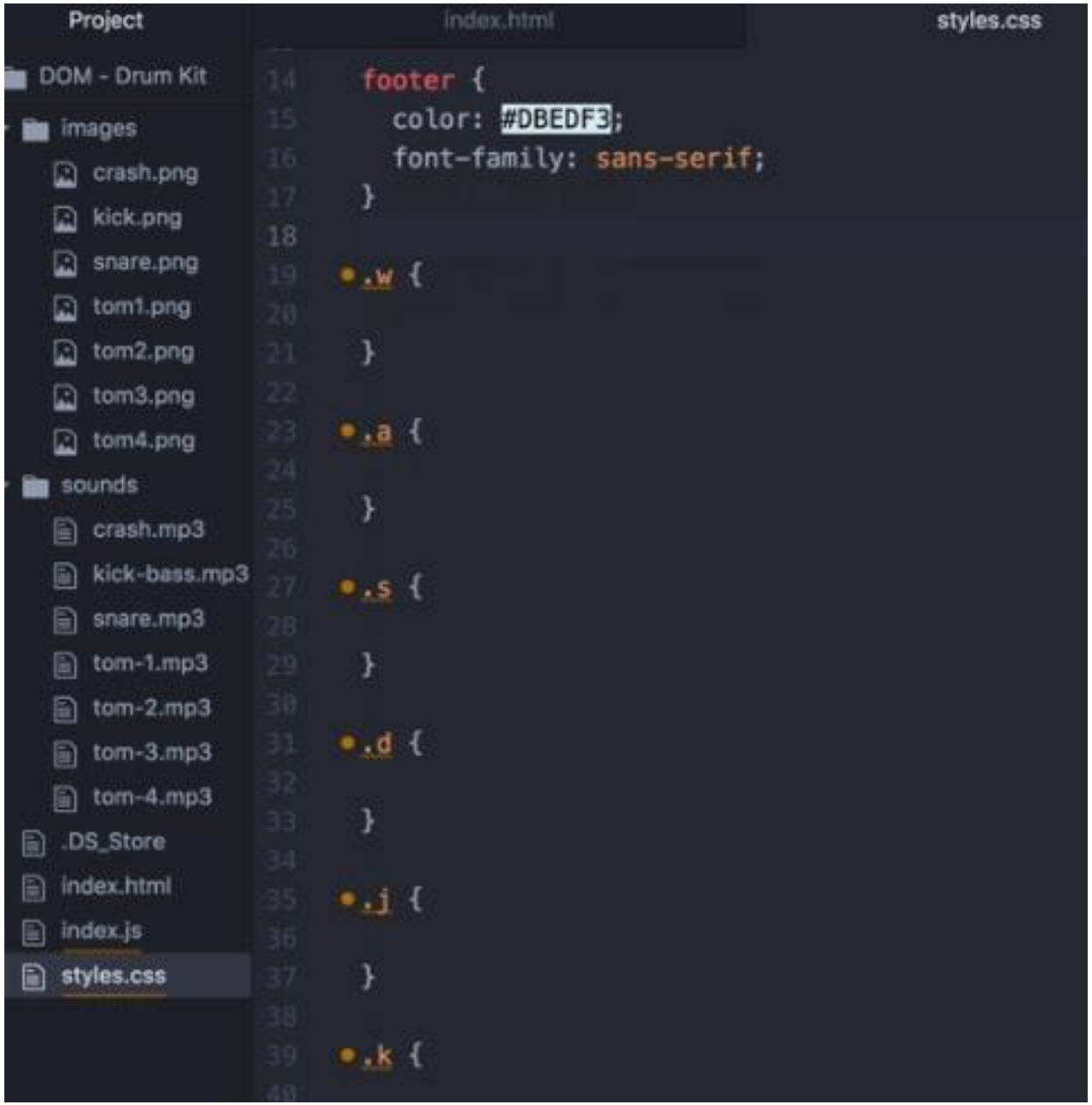
for (var i = 0; i<numberOfDrumButtons; i++) {

  document.querySelectorAll(".drum")[i].addEventListener("click", function () {

    var audio = new Audio("sounds/tom-1.mp3");
    audio.play();

  });

}
```


Project
index.html
styles.css
DOM - Drum Kit
images
crash.png
kick.png
snare.png
tom1.png
tom2.png
tom3.png
tom4.png
sounds
crash.mp3
kick-bass.mp3
snare.mp3
tom-1.mp3
tom-2.mp3
tom-3.mp3
tom-4.mp3
.DS_Store
index.html
index.js
styles.css
footer {
color: #DBEDF3;
font-family: sans-serif;
}
.w {
}
.a {
}
.s {
}
.d {
}
.j {
}
.k {

CSS

Drum Kit Starting Files > # styles.css > .l

```
15      color: #DBEDF3;
16      font-family: sans-serif;
17    }
18
19    .w {
20      background-image: url("images/tom1.png");
21    }
22
23    .a {
24      background-image: url("images/tom2.png");
25    }
26
27    .s {
28      background-image: url("images/tom3.png");
29    }
30
31    .d {
32      background-image: url("images/tom4.png");
33    }
34
35    .j {
36      background-image: url("images/snare.png");
37    }
38
39    .k {
40      background-image: url("images/crash.png");
41
42    }
43
44    .l {
45      background-image: url("images/kick.png");
46    }
47
```

Drum Kit









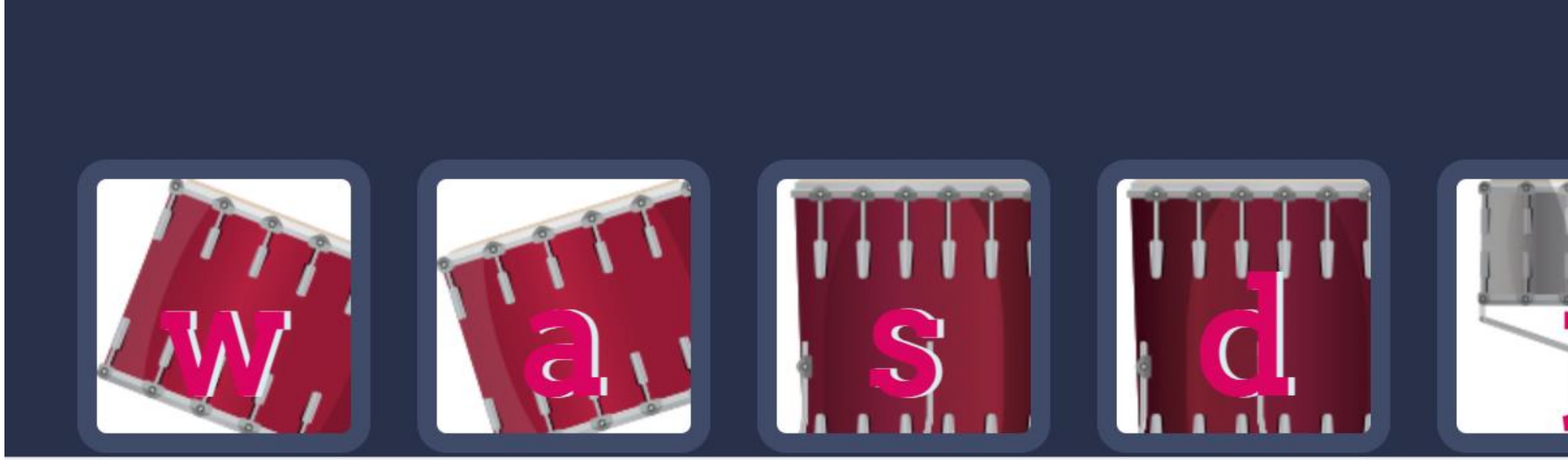






Made with ❤ in London.

this



Elements Console Sources Network Performance Memory Application Security Lighthouse Recorder Performance insights

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
▶ <head> … </head>
… ▼ <body> == $0
    <h1 id="title">Drum Kit</h1>
  ▶ <div class="set"> … </div>
    <script src="./index.js" charset="utf-8"></script>
    <footer> Made with ❤ in London. </footer>
  </body>
```

html body

Console What's New

top Filter

```
<button class="a drum">a</button>
<button class="w drum">w</button>
<button class="a drum">a</button>
<button class="s drum">s</button>
<button class="d drum">d</button>
```

>

```
1  var numberOfDrumButtons = document.querySelectorAll(".drum").length;
2
3  for(var i = 0;i< numberOfDrumButtons;i++)
4  document.querySelectorAll("button")[i].addEventListener("click", functi
5      // var audio = new Audio("sounds/tom-1.mp3");
6      // audio.play();
7      console.log(this);
8
9  });
10
11
```

```
var numberOfDrumButtons = document.querySelectorAll(".drum").length;

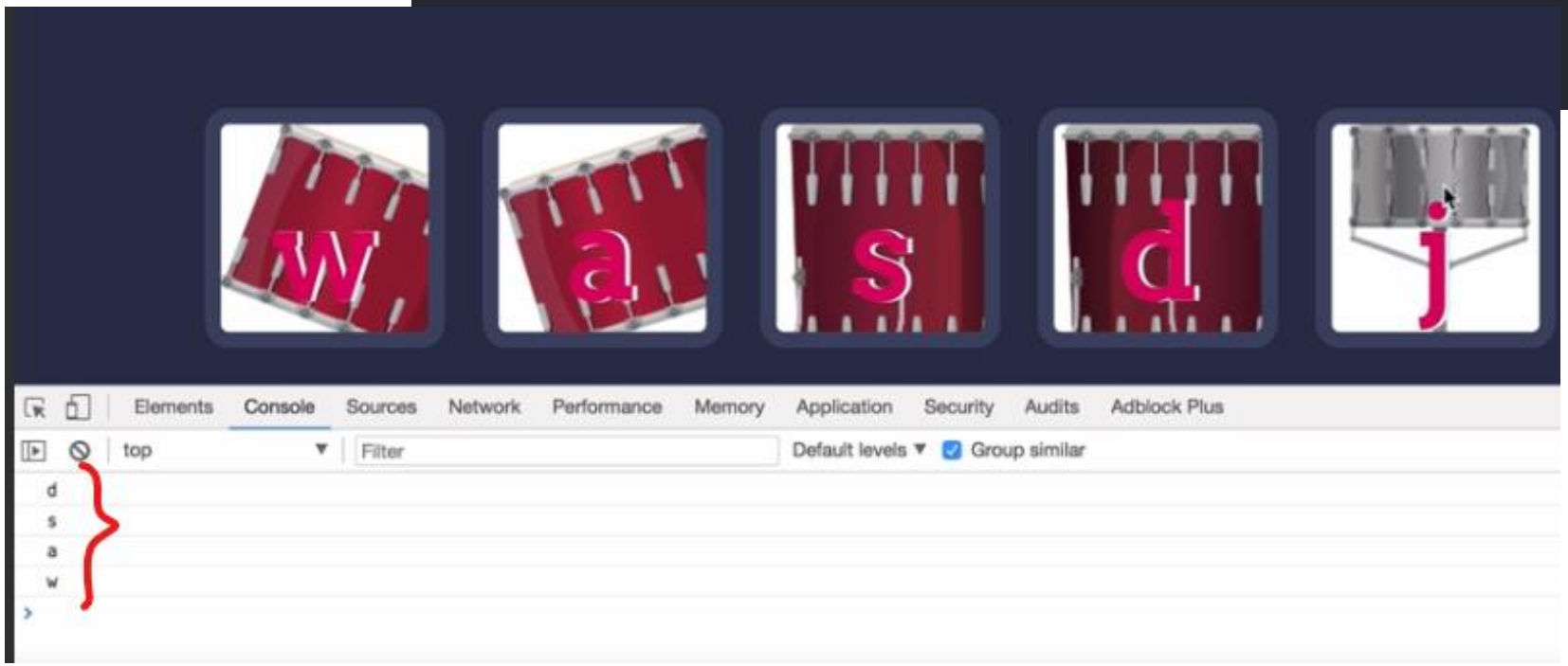
for (var i = 0; i<numberOfDrumButtons; i++) {

  document.querySelectorAll(".drum")[i].addEventListener("click", function () {

      console.log(this.innerHTML);

  });

}
```



White on Click

Drum Kit Starting Files > JS index.js > addEventListener("click") callback

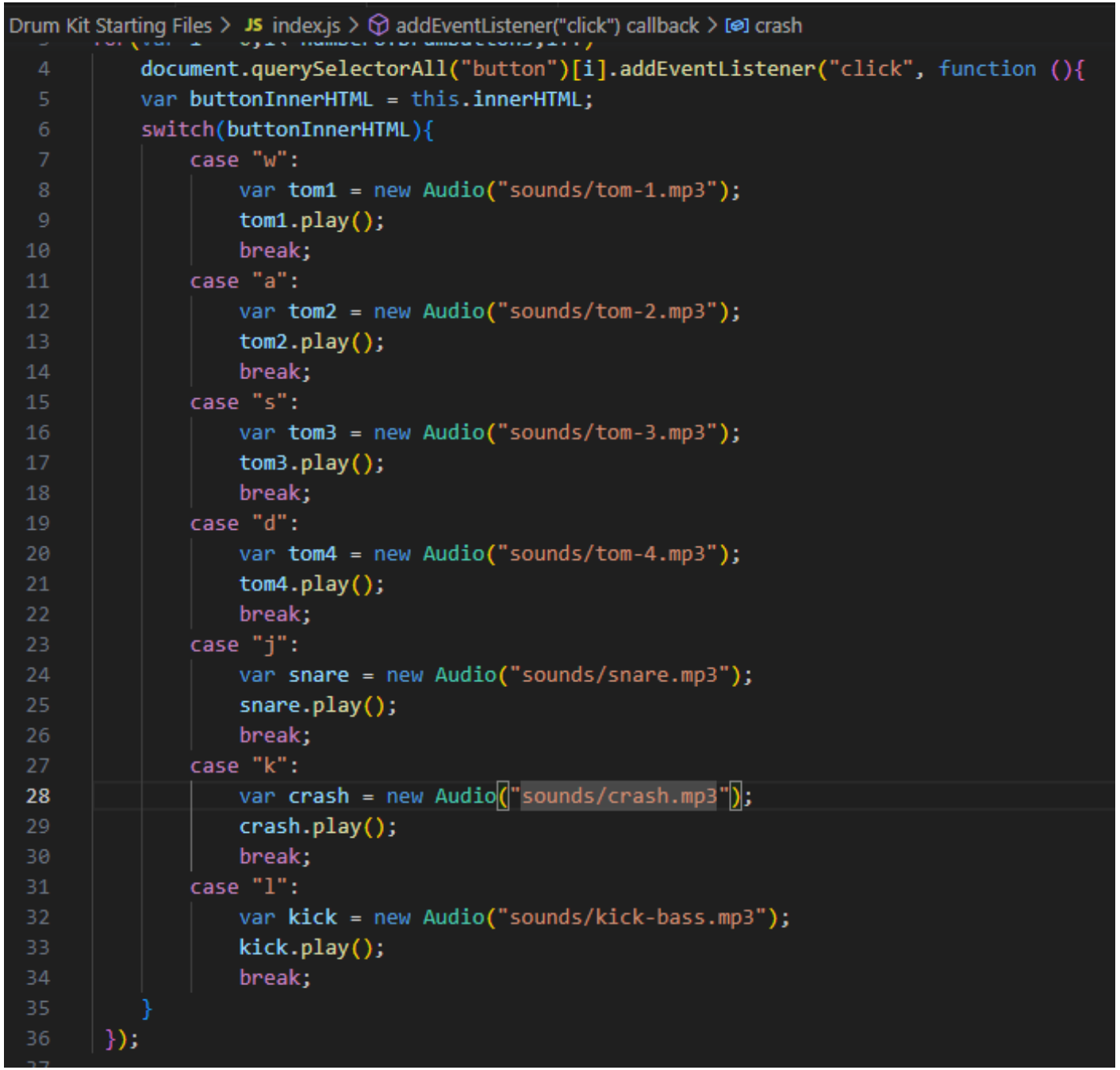
```
1   var numberOfDrumButtons = document.querySelectorAll(".drum").length;
2
3   for(var i = 0;i< numberOfDrumButtons;i++)
4   document.querySelectorAll("button")[i].addEventListener("click", function (){
5       // var audio = new Audio("sounds/tom-1.mp3");
6       // audio.play();
7       //console.log(this);
8       this.style.color = "white";
9
10  });
11
12
```

Switch statement in Javascript

Drum Kit Starting Files > JS index.js > addEventListener("click") callback

```
1   var numberOfDrumButtons = document.querySelectorAll(".drum").length;
2
3   for(var i = 0;i< numberOfDrumButtons;i++)
4       document.querySelectorAll("button")[i].addEventListener("click", function (){
5       var buttonInnerHTML = this.innerHTML;
6       switch(buttonInnerHTML){
7           case "w":
8               var audio = new Audio("sounds/tom-1.mp3");
9               audio.play();
10              break;
11
12      }
13  });
```

All dram


Drum Kit Starting Files > index.js > addEventListener("click") callback > crash
document.querySelectorAll("button")[i].addEventListener("click", function (){
var buttonInnerHTML = this.innerHTML;
switch(buttonInnerHTML){
case "w":
var tom1 = new Audio("sounds/tom-1.mp3");
tom1.play();
break;
case "a":
var tom2 = new Audio("sounds/tom-2.mp3");
tom2.play();
break;
case "s":
var tom3 = new Audio("sounds/tom-3.mp3");
tom3.play();
break;
case "d":
var tom4 = new Audio("sounds/tom-4.mp3");
tom4.play();
break;
case "j":
var snare = new Audio("sounds/snare.mp3");
snare.play();
break;
case "k":
var crash = new Audio("sounds/crash.mp3");
crash.play();
break;
case "l":
var kick = new Audio("sounds/kick-bass.mp3");
kick.play();
break;
}
});

Click to dram sound



JavaScript Objects

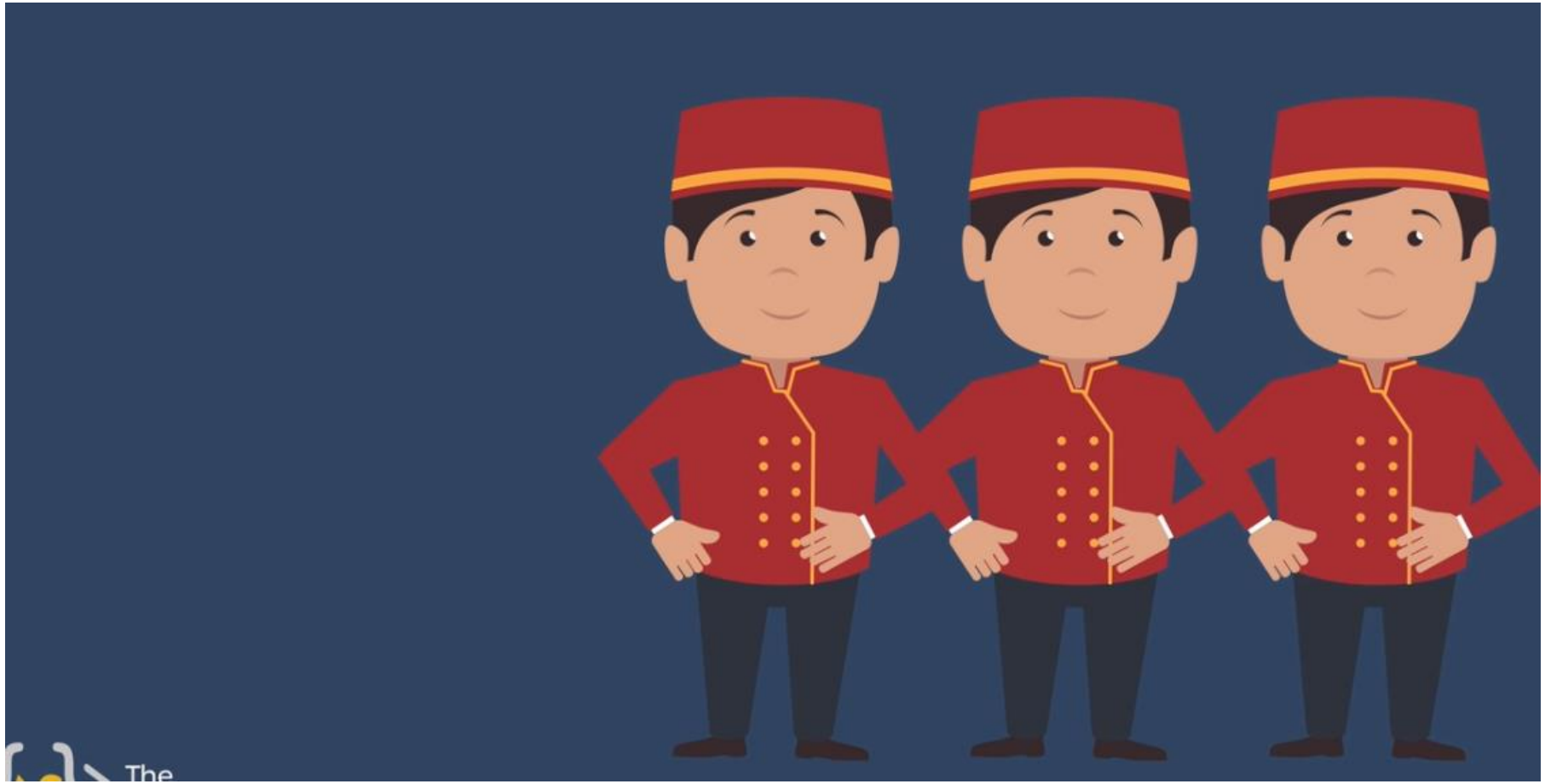
Dominating the Web, One Object at a Time.

Bell Boy 1

```
var bellBoy1Name = "Timmy";
var bellBoy1Age = 19;
var bellBoy1HasWorkPermit = true;
var bellBoy1Languages = [
    "French",
    "English"
]
```

```
alert("Hello, my name is " +
bellBoy1Name)
```




The

Properties	bellBoy1	bellBoy2	bellBoy3
name	“Timmy”	“Jimmy”	“Bitty”
age	19	21	18
hasWorkPermit	TRUE	FALSE	TRUE
languages	[“French”, “English”]	[“English”, “German”]	[“English”]



```
var bellBoy2 = {
   name: "Timothy",
   age: 19,
   hasWorkPermit: true,
   languages: ["French",
   "English"]
}
var bellBoy3 = {
   name: "Timothy",
   age: 19,
   hasWorkPermit: true,
   languages: ["French",
   "English"]
```



The

Object

```
var bellBoy1 = {
   name: "Timmy",
   age: 19,
   hasWorkPermit: true,
   languages: ["French", "English"]
}
```

```
alert("Hello, my name is " +
      bellBoy1.name )
```



Constructor Function

```
function BellBoy (name, age, hasWorkPermit, languages) {
  this.name = name;
  this.age = age;
  this.hasWorkPermit = hasWorkPermit;
  this.languages = languages;
}
```

Constructor Function

```
function BellBoy (name, age, hasWorkPermit, languages) {
  this.name = name;
  this.age = age;
  this.hasWorkPermit = hasWorkPermit;
  this.languages = languages;
}
```

Initialise Object

```
var bellBoy1 = new BellBoy("Timmy", 19, true, ["French", "English"]);
var bellBoy1 = new BellBoy("Jimmy", 21, false, ["English", "German"]);
```

•
•
•

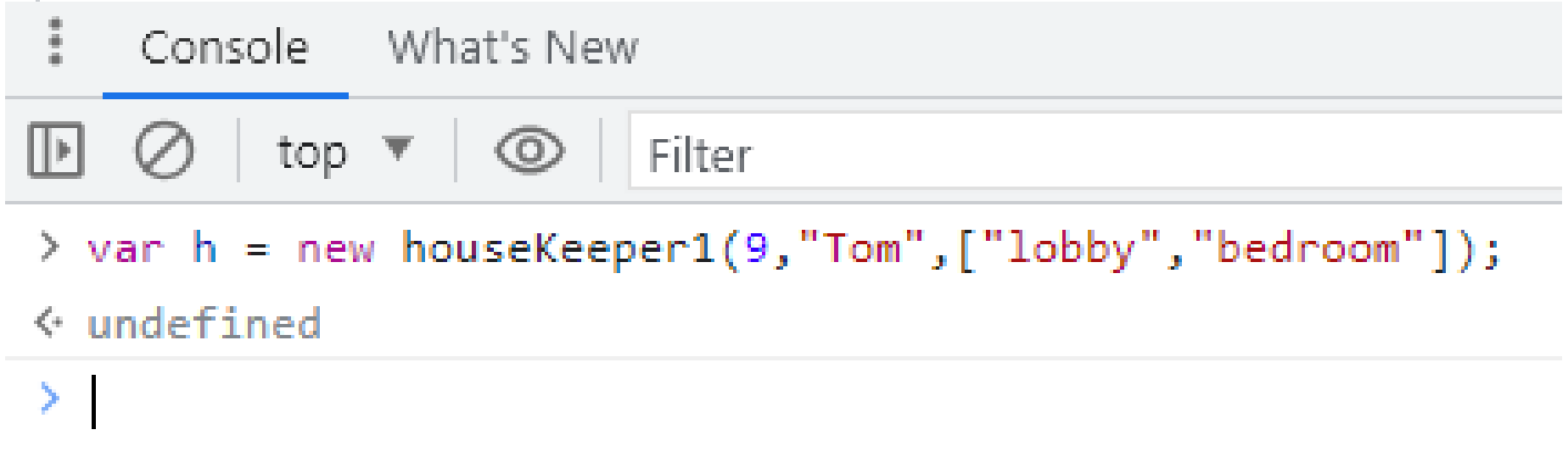
index.js* ×

```
var houseKeeper1 = {
    yearsOfExperience: 12,
    name: "Jane",
    cleaningRepertoire: ["bathroom", "lobby", "bedroom"]
}


var houseKeeper2 = {
    yearsOfExperience: ,
    name: "Jane",
    cleaningRepertoire: ["bathroom", "lobby", "bedroom"]
}
```

Creating class in javascript for initial object

```
1 function houseKeeper1(yearOfExperience,name,cleaningRepertoire){
2     this.yearOfExperience = yearOfExperience;
3     this.name= name;
4     this.cleaningRepertoire= cleaningRepertoire;
5
6 }
7
```



Methods

```
function moveSuitcase {
  alert("May I take your suitcase?");
  pickUpSuitcase();
  move();
}
```

```
var bellBoy1 = {
   name: “Timmy”,
   age: 19,
   hasWorkPermit: true,
   languages: [“French”, “English”],
   moveSuitcase: function() {
      alert(“May I take your suitcase?”);
      pickUpSuitcase();
      move();
   }
}
```

Call Method

```
bellBoy1.moveSuitcase();
```

Constructor Function

```
function BellBoy (name, age, hasWorkPermit, languages) {
  this.name = name;
  this.age = age;
  this.hasWorkPermit = hasWorkPermit;
  this.languages = languages;
  this.moveSuitcase = function() {
    alert("May I take your suitcase?");
    pickUpSuitcase();
     move();
   }
}
```

```
1  function houseKeeper1(yearOfExperience,name,cleaningRepertoire){
2      this.yearOfExperience = yearOfExperience;
3      this.name= name;
4      this.cleaningRepertoire= cleaningRepertoire;
5      this.clean = function(){
6          alert("Cleaning in progress...");
7      }
8  }
9
```

```
> var h = new houseKeeper1(9,"Tom",["lobby","bedroom"]);
< undefined
```

```
> h.clean();
< undefined
```

OOP of

```
var tom1 = new Audio("sounds/tom-1.mp3");
tom1.play();
```

Constructor Function

```
function Audio (fileLocation) {
  this.fileLocation = fileLocation;
  this.play = function() {
   //Tap into the audio hardware
   //Check the file at fileLocation exists
   //Check the file at fileLocation is a sound file
   //Play the file at fileLocation
  }
}
```

```
var tom1 = new Audio("sounds/tom-1.mp3");
tom1.play();
```

Using Keyboard Event Listeners

Clipboard Events

Event Name	Fired When
`cut`	The selection has been cut and copied to the clipboard
`copy`	The selection has been copied to the clipboard
`paste`	The item from the clipboard has been pasted

Keyboard Events

Event Name	Fired When
`keydown`	ANY key is pressed
`keypress`	ANY key except Shift, Fn, CapsLock is in pressed position. (Fired continou
`keyup`	ANY key is released

Mouse Events

Event Name	Fired When
`mouseenter`	A pointing device is moved onto the element that has the listener attached.

```
        kick.play();
        break;


      default: console.log(buttonInnerHTML);

    }

  });

}

addEventListener("keypress", function() {
  alert("Key was pressed!");
});
```



Example

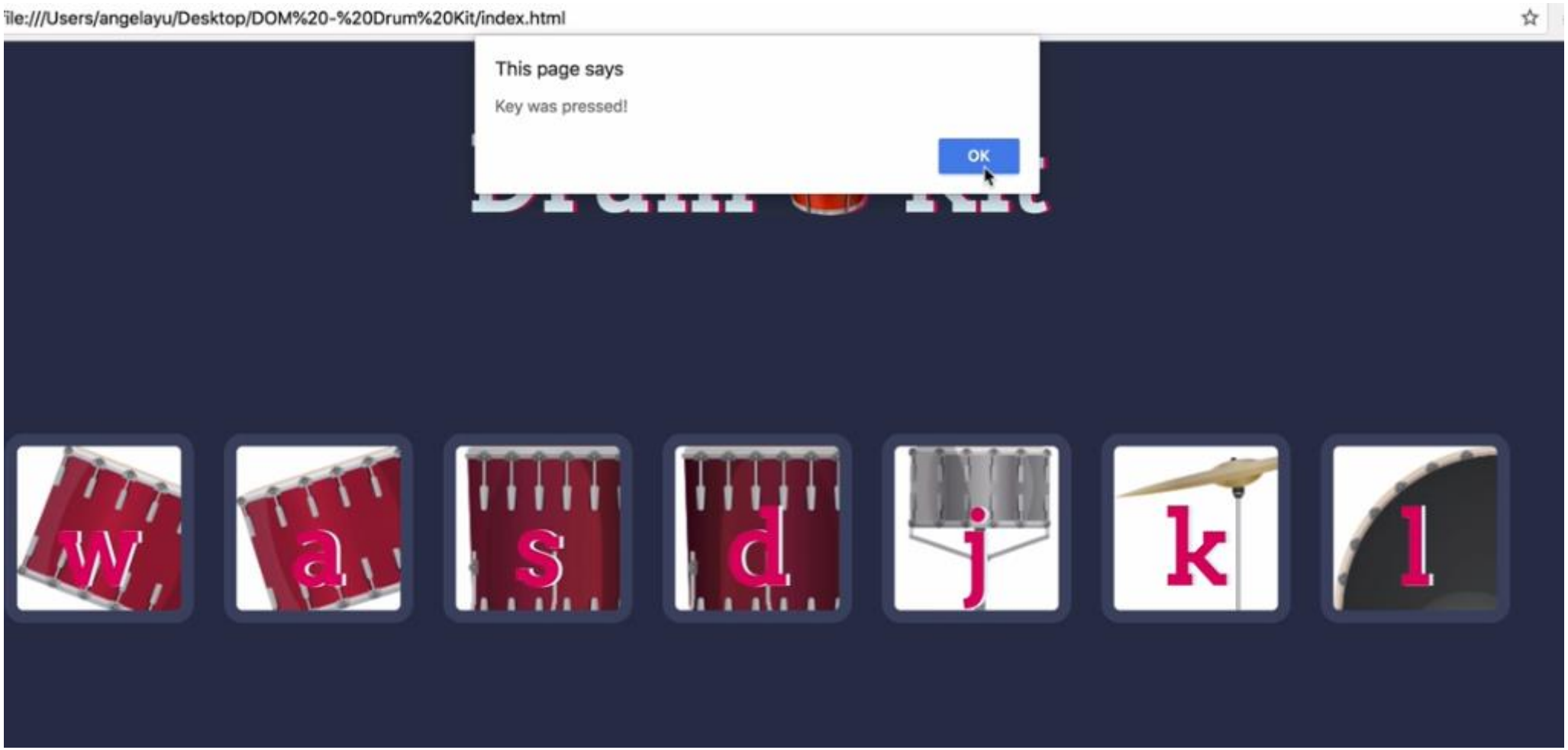
```
<!DOCTYPE html>
<html>
<head>
<script>

'use strict';

document.addEventListener('keypress', (event) => {
  const keyName = event.key;

  alert('keypress event\n\n' + 'key: ' + keyName);
});

</script>
</head>
<body>
</body>
</html>
```


ile:///Users/angelayu/Desktop/DOM%20-%20Drum%20Kit/index.html
This page says
Key was pressed!
OK
w
a
s
d
j
k
l

```
39
40  addEventListener("keypress",function(event){
41      console.log(event);
42  });
```

```
▼ KeyboardEvent {isTrusted: true, key: 'q', code: 'KeyQ', location: 0, ctrlKey: false, …}
    isTrusted: true
    altKey: false
    bubbles: true
    cancelBubble: false
    cancelable: true
    charCode: 113
    code: "KeyQ"
    composed: true
    ctrlKey: false
    currentTarget: null
    defaultPrevented: false
    detail: 0
    eventPhase: 0
    isComposing: false
    key: "q"
    keyCode: 113
    location: 0
    metaKey: false
    repeat: false
    returnValue: true
    shiftKey: false
  ▶ sourceCapabilities: InputDeviceCapabilities {firesTouchEvents: false}
  ▶ srcElement: body
  ▶ target: body
    timeStamp: 50042.799999952316
    type: "keypress"
  ▶ view: Window {window: Window, self: Window, document: document, name: '', location: Location, …}
    which: 113
  ▶ [[Prototype]]: KeyboardEvent
```

Event Click and Keypress

- Change old version to add new function inside.

Drum Kit Starting Files > JS index.js > ...

```
1  var numberOfDrumButtons = document.querySelectorAll(".drum").length;
2
3  for(var i = 0;i< numberOfDrumButtons;i++)
4      document.querySelectorAll("button")[i].addEventListener("click", function (){
5      var buttonInnerHTML = this.innerHTML;
6      makeSound(buttonInnerHTML);
7   });
8
9  document.addEventListener("keypress",function(event){
10     makeSound(event.key);
11 });
12
13 function makeSound(key){
14     switch(key){
15         case "w":
16             var tom1 = new Audio("sounds/tom-1.mp3");
17             tom1.play();
18             break;
19         case "a":
20             var tom2 = new Audio("sounds/tom-2.mp3");
21             tom2.play();
22             break;
23         case "s":
24             var tom3 = new Audio("sounds/tom-3.mp3");
25             tom3.play();
26             break;
27         case "d":
28             var tom4 = new Audio("sounds/tom-4.mp3");
29             tom4.play();
30             break;
31         case "j":
32             var snare = new Audio("sounds/snare.mp3");
33             snare.play();
34             break;
35         case "k":
36             var crash = new Audio("sounds/crash.mp3");
37             crash.play();
38             break;
39         case "l":
40             var kick = new Audio("sounds/kick-bass.mp3");
41             kick.play();
42             break;
43         default:
44             console.log(buttonInnerHTML);
45     }
46 }
```

Animation Button

```
        var crash = new Audio('sounds/crash.mp3');
        crash.play();
        break;

      case "l":
        var kick = new Audio('sounds/kick-bass.mp3');
        kick.play();
        break;


      default: console.log(buttonInnerHTML);

    }
  }


function buttonAnimation(currentKey) {

  var activeButton = document.querySelector("." + currentKey);



}
```

```
for (var i = 0; i < numberOfDrumButtons; i++) {

  document.querySelectorAll(".drum")[i].addEventListener("click", function() {

    var buttonInnerHTML = this.innerHTML;

    makeSound(buttonInnerHTML);

    buttonAnimation(buttonInnerHTML);

  });

}

document.addEventListener("keypress", function(event) {

  makeSound(event.key);

  buttonAnimation(event.key);

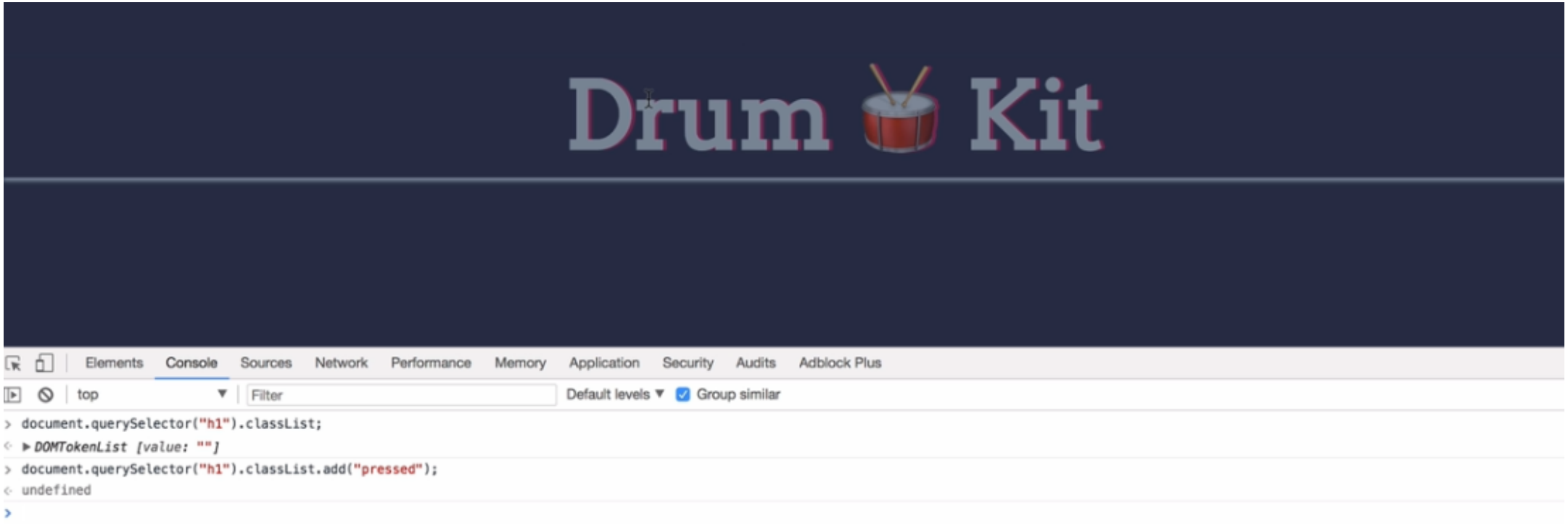
});


function makeSound(key) {

  switch (key) {
    case "w":
```


background-image: url("images/crash.png");
}
.l {
background-image: url("images/kick.png");
}
.set {
margin: 10% auto;
}
.pressed {
box-shadow: 0 3px 4px 0 #DBEDF3;
opacity: 0.5;
}
.red {
color: red;
}
.drum {
outline: none;
border: 10px solid #404B69;
font-size: 5rem;
font-family: 'Arvo', cursive;
line-height: 2;
font-weight: 900;
color: #DA0463;

Make a shadow to tag



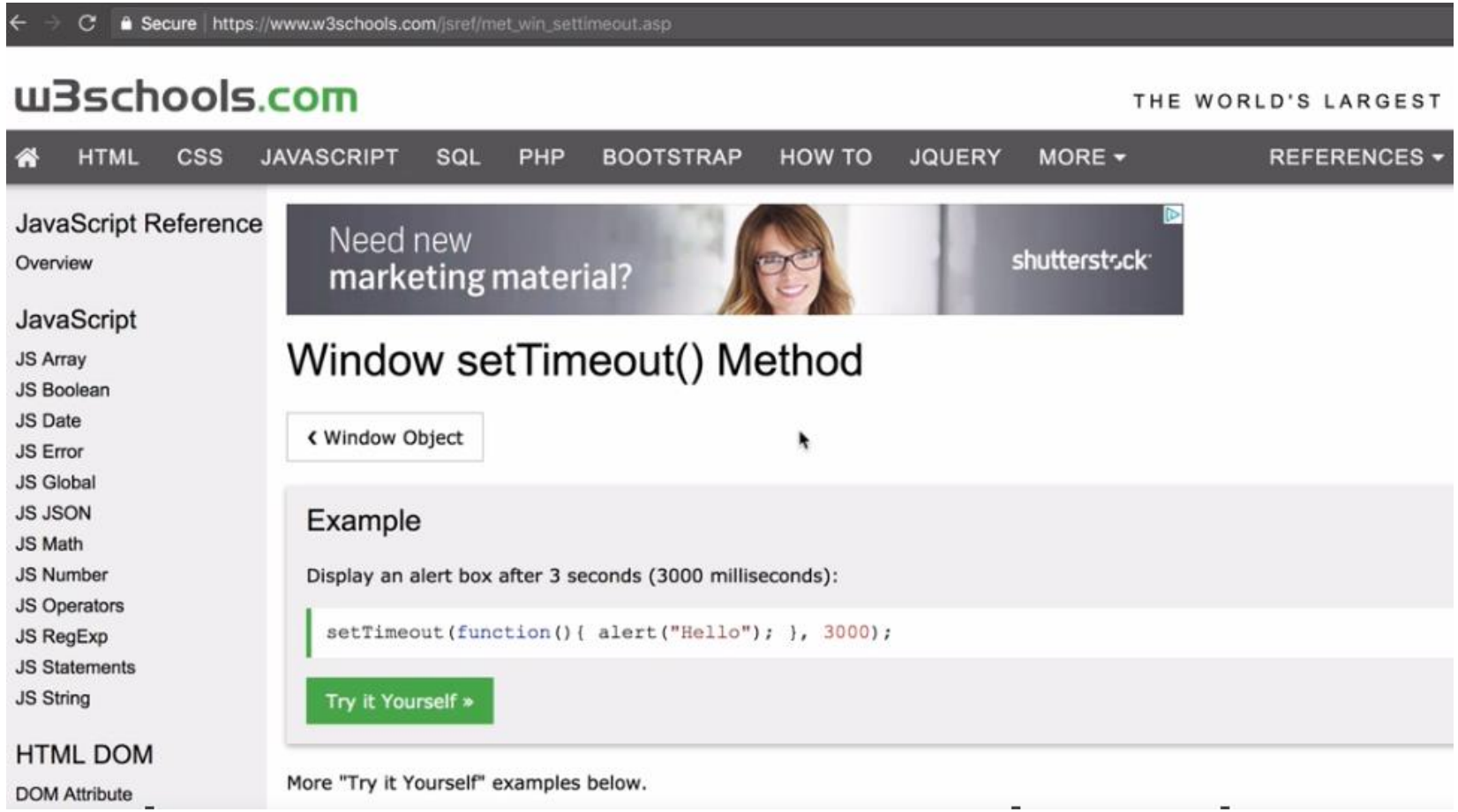
Add the class pressed to activeButton

```
function buttonAnimation(currentKey) {

  var activeButton = document.querySelector("." + currentKey);

  activeButton.classList.add("pressed");

}
```


Secure https://www.w3schools.com/jsref/met_win_settimeout.asp
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Window setTimeout() Method
‹ Window Object
Example
Display an alert box after 3 seconds (3000 milliseconds):
setTimeout(function(){ alert("Hello"); }, 3000);
Try it Yourself »
More "Try it Yourself" examples below.

setTimeout

```
function buttonAnimation(currentKey) {

  var activeButton = document.querySelector("." + currentKey);

  activeButton.classList.add("pressed");

  setTimeout(function() {
    activeButton.classList.remove("pressed");
  }, 100);

}
```

Try to click and keypress

